Self-Guided Laboratory Session
Image Enhancement & Morphological Image Processing

Duration: 2 Hours
Briefing: 10 min
Hands-on: 80–90 min
Evaluation: 20–30 min

Syllabus Reference

5. Image Enhancement [10 Hours]
    - Point processing techniques, histogram processing, equalization, contrast stretching, and thresholding
    - Spatial domain filtering, smoothing and sharpening filters, unsharp masking, and high-boost filtering
    - Frequency domain filtering, Fourier transform and its applications, filtering in the frequency domain

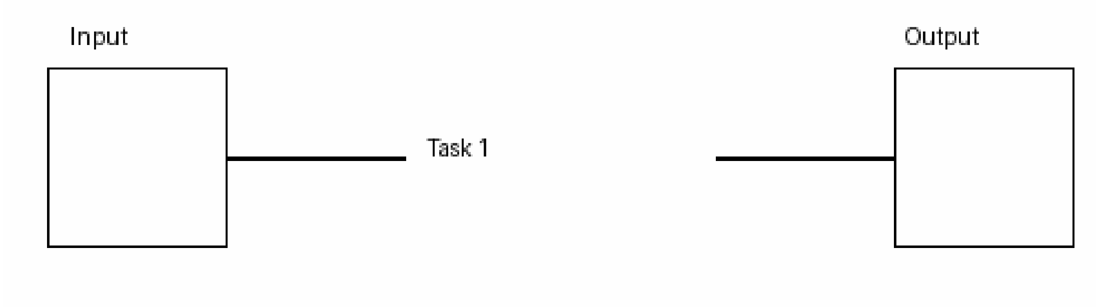
6. Image Restoration [9 Hours]
    - Image degradation and noise models, noise reduction techniques, spatial filtering for noise reduction
    - Inverse and Wiener filtering, motion blur, and its removal
    - Image registration techniques

7. Color Image Processing [4 Hours]
    - Color models (RGB, HSV, YCbCr)
    - Color transformations
    - Color image enhancement

8. Morphological Image Processing and Image Segmentation [5 Hours]
    - Basic morphological operations: dilation, erosion, opening, closing
    - Morphological algorithms for boundary extraction, region filling, and thinning
    - Thresholding techniques. Edge-based segmentation, Region-based segmentation

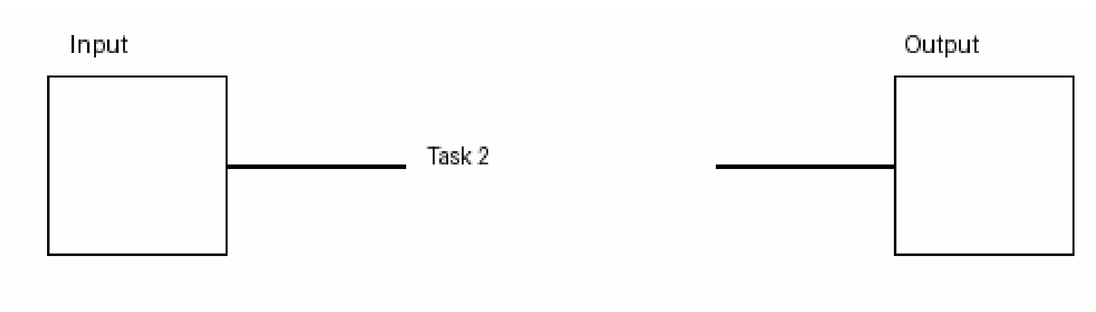
Task 1: Brightness & Contrast Adjustment

Increase/decrease brightness and contrast.
Deliverable: Original + Brightness +50, Brightness -50, Contrast ×1.5.



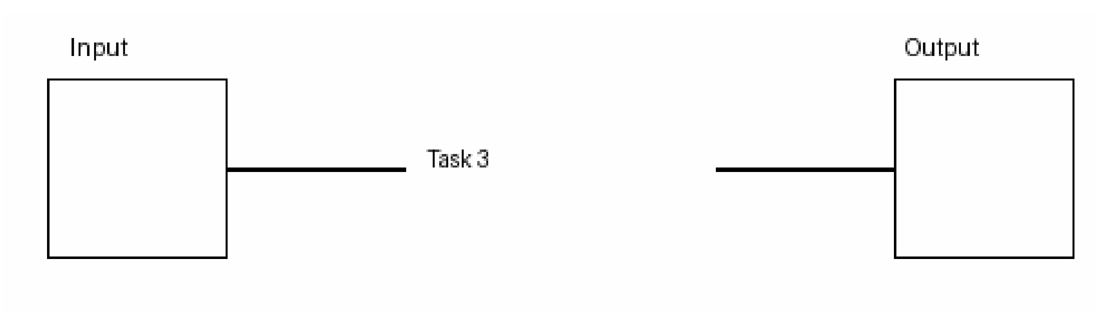
Task 2: Histogram Equalization

Apply histogram equalization and compare histogram before/after.



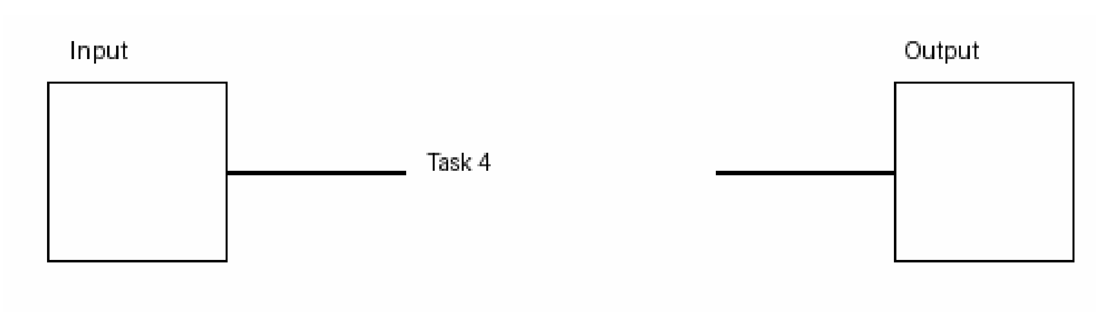
Task 3: Smoothing Filters

Compare Average, Gaussian and Median filters on noisy image.



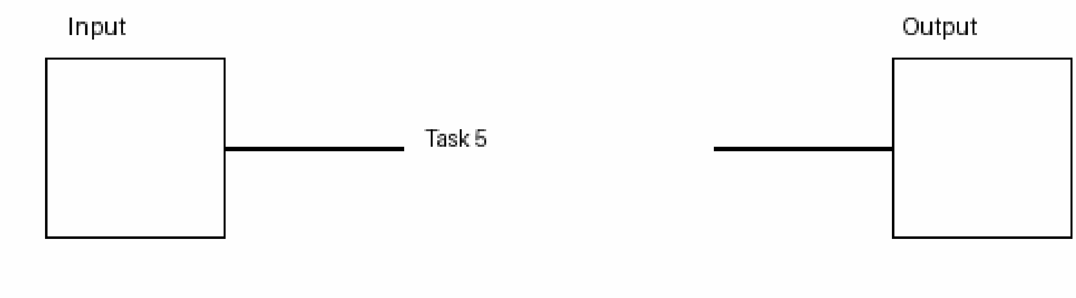
Task 4: Sharpening & Unsharp Masking

Create blurred image, mask and sharpened outputs.



Task 5: Morphological Processing & Segmentation

Threshold image, perform erosion, dilation, opening and closing; count objects.



Evaluation Rubric (10 Marks)

Criteria	Marks
Code runs correctly	3
Correct technique	3
Output quality	2
Explanation	2